

EVSP DR Current Research

Week of 14 September. Results checked 12 September 2026 at 06:26 EDT; overnight extensions are still running. Open [Figures with explanations](#) for timing, charging costs and geography, or [CG curves and bus schedules](#) for the restored figure library.

Current finding. With set covering and inherited columns, all six chains match the GIRO fleet at $k=3-6$. Chains 3, 5 and 6 also reach ten buses at $k=10$. The new 512-route import treatment reaches 11 buses at $k=11$ in chain 6; other completed extension cases need extra buses.

Integer fleet with full-pool inheritance

Each cell is the number of integer bus routes found. Green matches the target. Red requires extra buses. Dashes denote interrupted CG; see the C2 k10 note below.

Target buses	Chain 1	Chain 2	Chain 3	Chain 4	Chain 5	Chain 6
2	2	2	2	2	3	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	—	7	7	7	7	7
8	—	8	8	8	8	8
9	—	—	9	9	9	9
10	—	—	10	—	10	10

The table above retains the earlier full-pool treatment. The extension checks at most 512 inherited routes before CG; its new integer results are shown separately below. Runs continue through $k=15$ for all six chains.

$k=2$ starts from single-trip routes; later sizes reuse validated previous- k columns. Chain 3 $k=10$ also includes routes from a fresh solver solution. These are pool proofs. Chain 2 k10 (71 buses) had zero pricing iterations and no CG certificate. Individual route replay passed; duplicate-trip removal and shared charger capacity are not established by this table.

Fresh starts on the same chains

Each size starts independently. Both tables use covering and 240-kWh batteries with 240-kW charging. Fresh fleet proof status varies by cell.

Target buses	Chain 1	Chain 2	Chain 3	Chain 4	Chain 5	Chain 6
5	5	5	5	5	6	5
8	9	9	9	9	9	8
10	11	11	11	11	11	11
15	18	17	18	19	16	19

Inherited columns improve the available integer pool: chain 3 at $k=8$ improves from a proved nine-bus pool optimum to eight; chain 5 at $k=5$ improves from a proved six-bus pool optimum to five.

What is limiting progress

Checking saved routes took too long. Before starting CG on a larger trip set, we check that routes saved from the previous size are still valid. In chain 1 at $k=7$, chain 2 at $k=9$ and chain 4 at $k=10$, this checking used up the available time. We now check at most 512 saved routes, for at most 15 minutes, then let CG search for new routes. All three cases have now finished CG. Their integer solves found 8, 10 and 11 buses, respectively; only the first and third are proved minimums within their saved pools. Using fewer saved routes may affect the final integer solution, so this is a separate experiment.

Charger limits made the search for new routes very slow. In one three-bus test, the LP solved in 0.006 seconds, but one search for a new route took 7.15 hours. The run ran out of time before it could prove that no improving route remained. Its 16-bus integer solution uses only the routes found so far; it does not show that 16 buses are necessary. The timing figures explain these two different delays.

Case	Baseline	PARX 60 kW	Station capacity	Both
$k=1$, two duties tested	1 each	1 each	1 each	1 each
$k=2$, 23 trips	2	2	3	3
$k=3$, 35 trips	3	3	16	16

Sixteen is the result in a small generated pool, not a proved physical requirement.

Capacity-constrained schedules passed station sweeps. This pilot has no 65% return-SOC floor and is separate from the six baseline chains.

Overnight extension results

06:26 EDT: 58/87 CG certificates; 52 MIP results (22 warm, 30 component).

Warm starts check ≤ 512 saved routes. All CG results below are certified; times include preparation.

Chain and target	Buses	Pool fleet bound	Fleet proved in pool	CG minutes	Weighted LP objective
Chain 1, k=07	8	8	Yes	53.6	700313.457
Chain 1, k=08	9	9	Yes	49.5	800383.688
Chain 1, k=09	10	9	No	67.9	900453.496
Chain 1, k=10	11	11	Yes	91.2	1000511.205
Chain 2, k=09	10	9	No	64.0	900301.353
Chain 2, k=10	11	10	No	55.7	1000371.146
Chain 2, k=11	12	11	No	65.7	1100404.250
Chain 2, k=12	13	12	No	85.6	1200441.362
Chain 3, k=11	12	11	No	27.7	1100375.859
Chain 3, k=12	14	12	No	32.8	1200427.901
Chain 3, k=13	14	13	No	52.6	1300474.559
Chain 3, k=14	15	14	No	57.9	1400485.597
Chain 3, k=15	17	15	No	59.7	1500507.420
Chain 4, k=10	11	11	Yes	42.0	1000426.271
Chain 4, k=11	12	11	No	59.4	1100462.865
Chain 4, k=12	13	12	No	68.2	1200505.601
Chain 4, k=13	15	13	No	88.6	1300530.332
Chain 5, k=11	12	11	No	70.5	1100522.228
Chain 5, k=12	13	12	No	109.2	1200576.644

Chain 6, k=11	11	11	Yes	32.5	1100399.029
Chain 6, k=12	13	12	No	64.6	1200464.548
Chain 6, k=13	15	13	No	117.1	1300532.041

A pool fleet bound concerns saved columns only. When the proved fleet exceeds target, reaching target requires different columns. An unproved gap leaves both the integer search and column pool as possible limitations.

Decomposition of 32 duties

Grouping	Four component fleets	Combined buses	Target
join01	9 + 9 + 9 + 8	35	32
join02	9 + 9 + 9 + 8	35	32
join03	9 + 9 + 8 + 10	36	32
join08	9 + 9 + 10 + 9	37	32
join09	8 + 9 + 9 + 9	35	32

These combine four disjoint trip groups selected using GIRO duty membership. Original GIRO route columns were not supplied. Individual component routes passed replay; duplicate removal and shared charger capacity are not certified. These are feasible covering constructions under the baseline model, not proofs of the minimum fleet.

First completed speed comparisons

Case	Reference minutes	Optimized minutes	Observation
Fresh d00_g0	22.3	23.6	5.8% slower
Fresh d00_g1	52.2	52.1	Essentially unchanged
Warm C1 k8 reverse order	31.1	24.1	22.4% less time
Warm C1 k8 reference first	41.1	34.4	16.2% less time

All completed pairs reached matching certified LP objectives. The two warm comparisons cut import checking from 366/447 seconds to 5.6/6.8 seconds. Total time fell 22.4%/16.2%,

excluding common graph-cache preparation. CG iterations differed in each pair; this does not isolate the total speedup or establish an integer improvement.

Capacity speed tests: first arms for k1, k2 and the combined k1 peak12 case reached the pricing deadline without certificates. Their counterparts are running, so no paired speedup conclusion is available.

Default MIPs: 75 earlier and 52 extension attempts completed with zero confirmed preemptions. Three extension MIPs were running at collection.

Methods and evidence. CG route cost = 100,000 + electricity + 5 per charging start. MIP stage 1 minimizes buses. Stage 2 constrains buses $\leq$ the validated stage-1 incumbent and minimizes electricity plus start fees.

A pricing certificate establishes the LP result only in its represented graph and tolerance. A MIP proof concerns its saved columns. Fleet attainment, physical replay and shared-capacity validation are separate checks.

[Experiment register](#) · [Dated results and source tables](#) · [Experiment settings](#) · [Historical document and figures](#) · [Local prototype tests and charging-cost bug](#)

Storage, 12 September: 156 cold files archived losslessly; 133.74 GB freed. Active research and V2G data preserved. [Restore instructions](#).

Blocked decomposition case d00_g3: the original run timed out before CG, while constructing the graph. Its MIP remains blocked. A separate diagnostic crashed after 366 seconds (SIGSEGV), before its watchdog limit. Partial samples show JSON serialization while adding charging arcs; A local test also hung with the timer sampler and no solver, so this sampling method is being retired. That does not yet explain the cluster crash. [Diagnostic evidence](#). No long retry has been launched.